

1 Part 1: Contextualizing the Data

Let's try to understand the background of our dataset before diving into a full-scale analysis.

1.1 Question 1a

Based on the columns present in this data set and the values that they take, what do you think each row represents? That is, what is the granularity of this data set?

Each row of the data set represents a specific property. And each property will have data that represents that property in the column data.

1.2 Question 1b

Why do you think this data was collected? For what purposes? By whom?

This question calls for your speculation and is looking for thoughtfulness, not correctness.

I think that this data could've been put together by people who work in real estate, and want to know more about what goes into the values of properties. Or it could also be from official records by the local government.

1.3 Question 1c

Craft at least two questions about housing in Cook County that can be answered with this data set and provide the type of analytical tool you would use to answer it (e.g. “I would create a ____ plot of ____ and ____” *or* “***I would calculate the*** [summary statistic] for ____ and ____”). Be sure to reference the columns that you would use and any additional data sets you would need to answer that question.

I would like to know how important the land square feet is when we are determining sales price. I think generally the more land you have the higher the property value, but I would like to find this correlation through the use of a residuals plot. I also would like to know how important the age of a property is when determining its sales price. I would assume that the older a property is the lower it is valued, but I would like to compare properties of similar attributes but of different years in order to find this connection. I would create a scatter plot of age and sales price in order to see what this connection looks like.

1.4 Question 1d

Suppose now, in addition to the information already contained in the dataset, you also have access to several new columns containing demographic data about the owner, including race/ethnicity, gender, age, annual income, and occupation. Provide one new question about housing in Cook County that can be answered using at least one column of demographic data and at least one column of existing data and provide the type of analytical tool you would use to answer it.

I would like to see if there is any connection between their age and what type of porch that they have. For each type of porch I would create a violin plot to visualize if there is a greater density of older homeowners who have a porch.

1.5 Question 2a

Using the plots above and the descriptive statistics from `training_data['Sale Price'].describe()` in the cells above, identify one issue with the visualization above and briefly describe one way to overcome it.

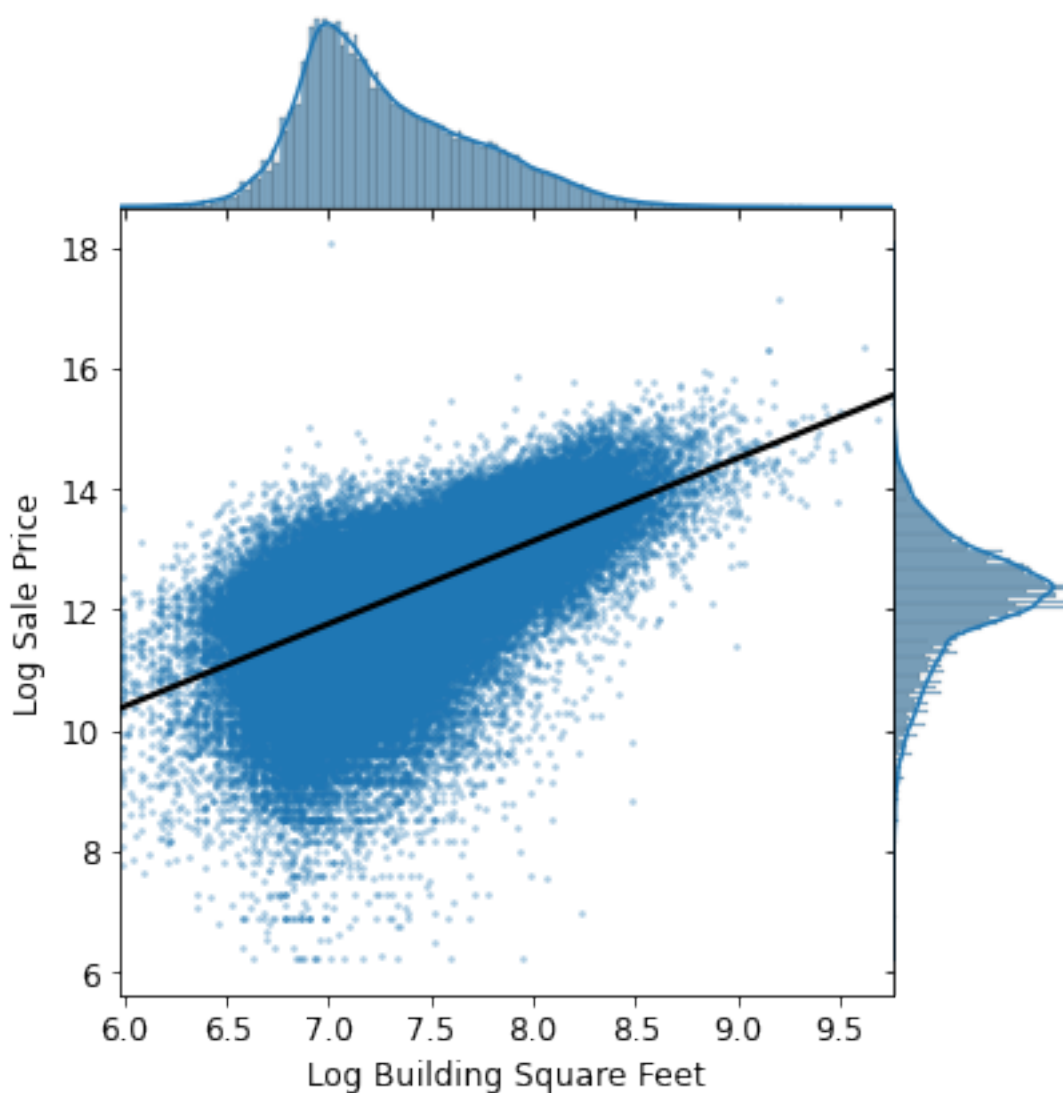
The x axis for both graphs squish the graphs to where it is no very clear what trends we are seeing within the data. We also do not have a y axis label, so if we were not able to look at the code it wouldn't be clear what the sale price was being connected to. We could overcome this by adding a y-label, along with changing the bounds of the x axis.

1.6 Question 3c

In the visualization below, we created a `jointplot` with `Log Building Square Feet` on the x-axis, and `Log Sale Price` on the y-axis. In addition, we fit a simple linear regression line through the bivariate scatter plot in the middle.

Based on the following plot, would `Log Building Square Feet` make a good candidate as one of the features for our model? Why or why not?

Hint: To help answer this question, ask yourself: what kind of relationship does a “good” feature share with the target variable we aim to predict?



Yes I would say that “Log Building Square Feet” makes a good candidate as one of the features for our model. This is because we do see a positive linear relationship when compared to the “Log Sale Price”. This means we have a feature that can help us predict the sale price from the square footage of the building.

1.7 Question 5c

Create a visualization that clearly and succinctly shows if there exists an association between **Bedrooms** and **Log Sale Price**. A good visualization should satisfy the following requirements: - It should avoid overplotting. - It should have clearly labeled axes and a succinct title. - It should convey the strength of the correlation between **Sale Price** and the number of rooms: in other words, you should be able to look at the plot and describe the general relationship between **Log Sale Price** and **Bedrooms**

Hint: A direct scatter plot of the **Sale Price** against the number of rooms for all of the households in our training data would result in overplotting (since there are only a small discrete number of bedrooms) - so **don't use a scatter plot**.

```
In [225]: sns.boxplot(data = training_data, x = "Bedrooms", y = "Log Sale Price")  
          plt.title("Comparing the Number of Bedrooms to the Log Sale Price")
```

```
Out[225]: Text(0.5, 1.0, 'Comparing the Number of Bedrooms to the Log Sale Price')
```

